S4 Experimental NMR details and ^{15}N solid-state NMR spectra **S.6**

S5 g_R model with no averaging across layers **S.6**

S6 Structural models and input files **S.9**

References **S.10**

S1 Comment on quantitative vs. non-quantitative experimental spectra

CPMAS does not provide fully quantitative ^{13}C NMR spectra. This is inherent to the method as the efficiency of polarization transfer depends on the strength of the heteronuclear dipolar interaction, which is generally different for different chemical groups. Transferring nuclear spin polarization from ^{1}H nuclei to ^{13}C nuclei typically is more efficient for carbon atoms bound to hydrogen atoms, as compared to carbon atoms further away from ^{1}H nuclei. Therefore, the contribution of lineshapes in the spectrum that are caused by hydrogenated carbon sites is larger than non-hydrogenated carbon sites, resulting in relative integrals of lineshapes that do not reflect the actual relative populations of these carbon atoms in the system (opposite to direct excitation spectra which are quantitative in nature). Initially, ^{13}C single pulse direct excitation experiments were also performed in order to obtain quantitative spectra. However, these were extremely time consuming, and, more importantly, suffered from very large background noise. As such, it was impossible to obtain usable quantitative ^{13}C NMR spectra. Nevertheless, with our computational models, we produce the underlying quantitative spectra and fit to the non-quantitative spectrum, because we feel that showing the quantitative computational model spectrum carries more value than varying the individual integrals of the constituting lineshapes and as such showing a non-quantitative computational spectrum. We note that the fitting procedure is only done to obtain (i) the overall amplitude